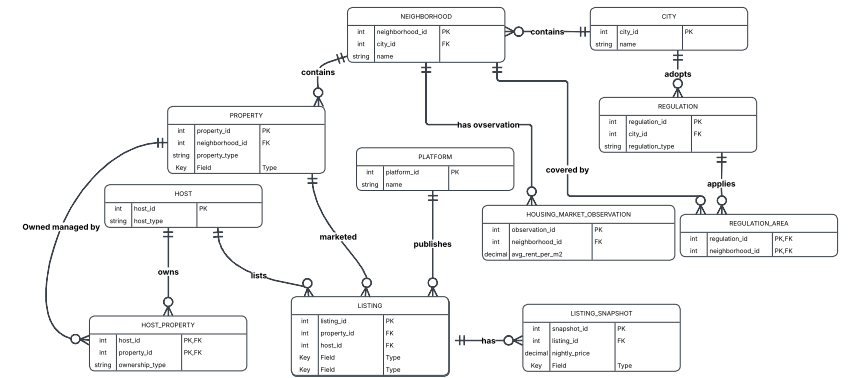
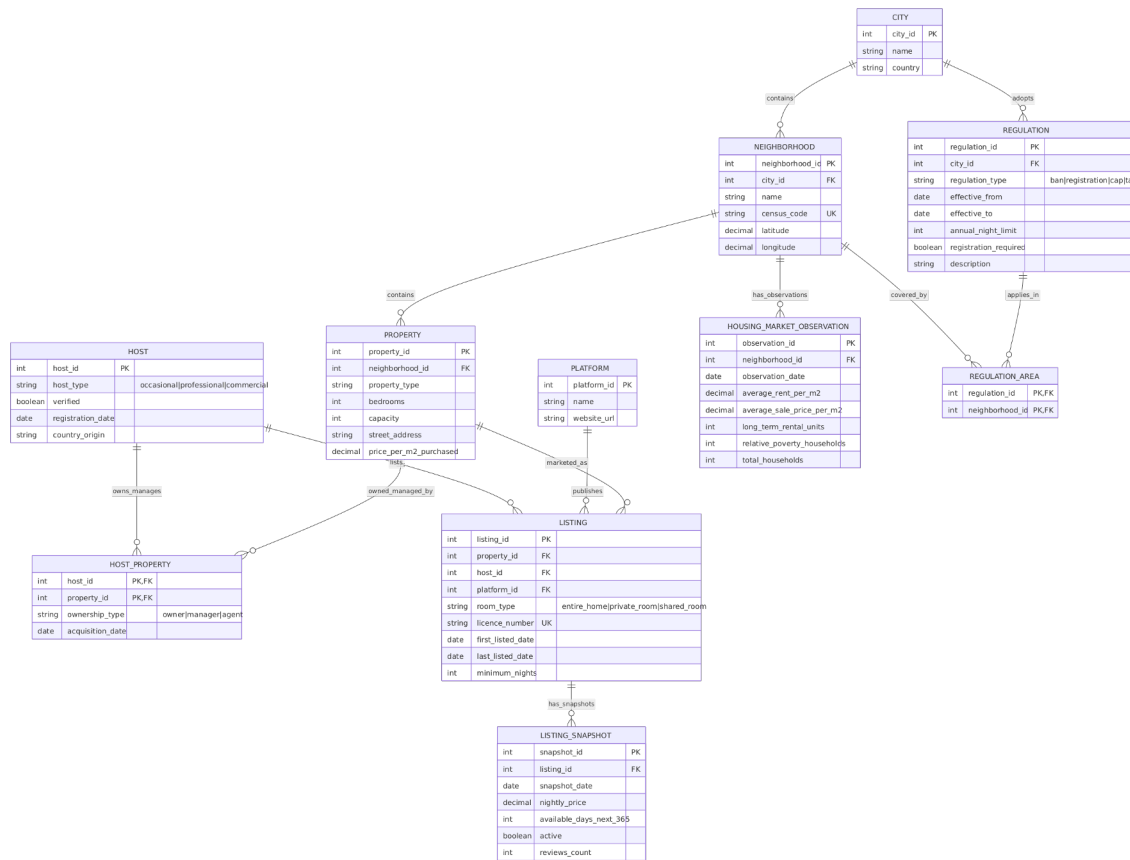




Databases KEN2110

Data Modelling

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Unnormalized State (Initial Problem)

Original Table: AIRBNB_RAW_DATA

This represents how data might be stored initially without normalization:

listing_id	property_id	host_id	host_name	city_name	regulation_type	neighborhood_name	room_type	price
L001	P001	H001	Jean	Barcelona	ban	Gothic Quarter	entire_home	85
L002	P002	H002	Maria	Barcelona	ban	Eixample	private_room	45
L003	P001	H001	Jean	Barcelona	ban	Gothic Quarter	private_room	40

Problems with Unnormalized Data:

1. Data Redundancy - Repeated values like 'Barcelona', 'Spain', and 'Jean', 'France' appear multiple times
2. Update Anomalies - Changing Jean's country requires updating 3+ rows; risk of inconsistency
3. Insertion Anomalies - Cannot add a regulation without a corresponding listing
4. Deletion Anomalies - Deleting all listings for Jean loses information that he was a host

First Normal Form (1NF)

Atomic Values Only - No Multi-valued Attributes

Goal: Ensure each cell contains only atomic (indivisible) values. Separate entities by concern.

Table 1: CITY (After 1NF)

city_id (PK)	city_name	country
C001	Barcelona	Spain
C002	Berlin	Germany

Table 2: HOST (After 1NF)

host_id (PK)	host_name	host_country
H001	Jean	France
H002	Maria	Spain

Table 3: NEIGHBORHOOD (After 1NF)

neighborhood_id (PK)	city_id (FK)	neighborhood_name
N001	C001	Gothic Quarter
N002	C001	Eixample

1NF Achieved

- All values are atomic
- Each entity has its own table
- Foreign keys establish relationships

Second Normal Form (2NF)

Remove Partial Dependencies

Goal: All non-key attributes must depend on the ENTIRE primary key, not just part of it.

Problem: LISTING_SNAPSHOT - Before 2NF

Partial dependency found: property_type and bedrooms depend only on listing_id, not on (listing_id, snapshot_date)

listing_id (PK)	snapshot_date (PK)	nightly_price	available	property_type	bedrooms
L001	2024-07-02	85	300	apartment	2
L001	2024-07-03	85	299	apartment	2

Solution: After 2NF - Extract into Separate Tables

Table A: LISTING_SNAPSHOT (only temporal data)

listing_id (FK)	snapshot_date (PK)	nightly_price	available_days	active
L001	2024-07-02	85	300	TRUE
L001	2024-07-03	85	299	TRUE

Table B: PROPERTY (static attributes remain separate)

property_id (PK)	neighborhood_id (FK)	property_type	bedrooms	capacity
P001	N001	apartment	2	4

2NF Achieved

- All partial dependencies removed
- Composite keys only reference attributes that depend on all key components

Third Normal Form (3NF)

Remove Transitive Dependencies

Goal: Non-key attributes must NOT depend on other non-key attributes. They must depend only on the primary key.

Problem: NEIGHBORHOOD - Before 3NF

Transitive dependency found: neighborhood_id → city_id → city_name

neighborhood_id (PK)	city_id (FK)	neighborhood_name	city_name	country
N001	C001	Gothic Quarter	Barcelona	Spain

Solution: After 3NF - Extract City Attributes to CITY Table

Table A: CITY

city_id (PK)	city_name	country
C001	Barcelona	Spain

Table B: NEIGHBORHOOD

neighborhood_id (PK)	city_id (FK)	neighborhood_name
N001	C001	Gothic Quarter

3NF Achieved

- All transitive dependencies removed
- Non-key attributes depend only on the primary key
- Single source of truth for each entity

Conclusion

The database schema has been normalized through 1NF, 2NF, and 3NF:

1. 1NF: Separated mixed entities into atomic tables with dedicated relationships
2. 2NF: Removed partial dependencies from composite keys
3. 3NF: Removed transitive dependencies between non-key attributes

The result is a robust, maintainable schema that prevents anomalies (update, insertion, deletion) and supports efficient querying for the Airbnb housing market analysis.